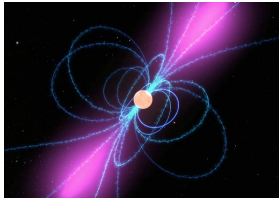


A Quantum Hunt for Lighthouses in the Cosmic Dark

Background

Finding pulsars in modern radio survey data is a classification problem at massive scale – surveys generate far more candidate signals than are able to be manually inspected, and actual pulsars are a small minority of all the detections. To automate classification, we use a Variational Quantum Classifier: a quantum feature map encodes classical data into a quantum state, and a trainable ansatz circuit is optimized to separate pulsar from non-pulsar, measured through a chosen entanglement structure. We tested two backend configurations – (1) the AER simulator with the COBYLA optimizer, and (2) the paper's own default sampler with the SLSQP optimizer, for direct comparison. Models were trained on 300 samples and tested on the full remaining ~17,600 sample pool. The dataset we use is HTRU2 (High Time Resolution Universe 2) which contains 17,898 candidate radio signals, where 9% are confirmed pulsars. The rest is noise. In our VQC, we focus our runs on



Above: Illustration of Pulsar Clouds of charged particles move along the magnetic field lines (blue) and create a lighthouse-like beam of gamma rays (purple)
Credit: NASA

Metrics

We report Accuracy, Precision, Recall, F1, False Positive Rate (FPR), and Matthews Correlation Coefficient (MCC). With only ~9% of the data being pulsars, a model that only predicts that a datapoint is not a pulsar will have 91% accuracy, despite being a useless model. Because of this, we prioritize MCC and F1, which capture performance on the rare, minority pulsar class specifically. A positive (P) is a datapoint marked as a pulsar. True/false means whether that call was correct. Accuracy, in layman's terms is how right the model is overall. Precision means, of everything flagged as a pulsar, how many actually were. Recall is of all real pulsars, how many were caught. F1 is only high if precision and recall are high. MCC is the best benchmark of how well the model performs in all four categories.

Metric	Formula
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$
Precision	$\frac{TP}{TP + FP}$
Recall	$\frac{TP}{TP + FN}$
F1	$2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
FPR	$\frac{FP}{FP + TN}$
MCC	$\frac{(TP \cdot TN - FP \cdot FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$

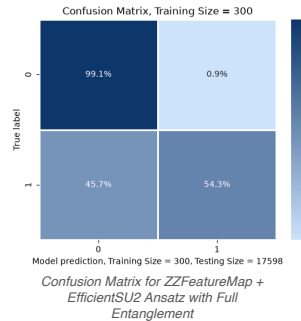
Left: Table containing each metric and its formula, where $TP/TN/FP/FN = \text{True/False Positive/Negative}$

The universe's most precise clocks are buried in radio noise. We use Quantum Machine Learning to dig them out.

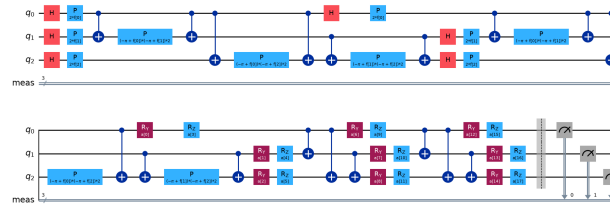
Pulsars are spinning neutron stars that sweep beams of radio waves through space like a lighthouse. They allow scientists to test general relativity, observe matter at densities impossible to recreate on Earth, and even help to detect gravitational waves. So how can we help find them?

Our Results

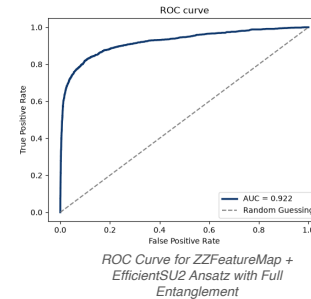
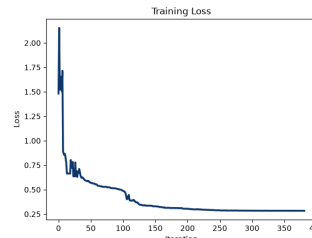
ZZFeatureMap and PauliFeatureMap produce an identical circuit with default parameters (Pauli defaults to ZZ encoding). Souza et al.'s own methodology never customized Pauli's term set either, but since we are using seeds to ensure our runs are fully reproducible, we decided to drop PauliFeatureMap as it's redundant. Holding that fixed, we swept all 9 ansatz x entanglement combinations across 5 independently-seeded runs. EfficientSU2 with full entanglement consistently scored the highest by MCC, which matches Souza et al.'s own configuration. We trained each run on 300 samples and tested on the remaining ~17,600.



The graphs above and to the right, from one representative seed, show this configuration's confusion matrix, training loss curve, ROC curve with AUC, and circuit diagram. The loss curve tracks cross-entropy loss as the optimizer adjusts the model's parameters during training. A downward trend shows learning, and a flattening curve shows convergence. The ROC curve plots true positive rate against false positive rate across every possible classification threshold, illustrating the core tradeoff between catching more real pulsars and flagging more false positives. AUC summarizes this into a single number (where 1 is perfect separation and 0.5 is random guessing). The confusion matrix fixes one threshold and shows the resulting classifications. The top-left/bottom-right are correct non-pulsar/pulsar calls, and the off-diagonal cells are the models' two failure modes; false pulsar flags are the top-right and missed pulsars bottom-left.



Above: 3-Qubit Quantum Circuit showcasing ZZFeatureMap + EfficientSU2 Ansatz with Full Entanglement

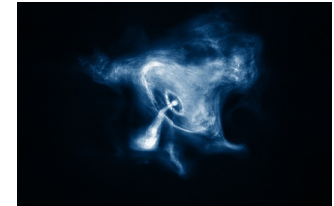


Conclusion and Outlook

Severe class imbalance (~9% pulsars) biases models towards the majority class. All results are simulation-only. The results are seed sensitive. AER + COBYLA includes real finite-shot statistical noise (4,096 shots per evaluation), whereas the SLSQP config's default sampler runs in exact mode with no sampling at all. Souza et al. report MCC 0.670 with 300 training samples for ZZFeatureMap + EfficientSU2 + full entanglement. Our replication closely matched the paper's reported best config in overall MCC, but precision/recall patterns were inverted. We ruled out seed variance as a cause via a multi-sweep. The most probable explanation is the substantial software version gap between the paper's implementation and ours. However, conservative, majority-class-learning predictions when dealing with class imbalance are also commonly reported for these models elsewhere in the VQC literature.

Future Plans

We're continuing this research through Fall 2026 with Souza et al.'s authors. Next steps: extend results to IonQ hardware simulation (most current pulsar-classifications research uses simulators only) and run our own classical ML baselines directly to compare to our QML method.



Above: X-Ray image of the Crab Nebula, provided by the Chandra observatory.
Credit: X-ray: NASA/CXC/SAO; Optical: NASA/STScI; Infrared: NASA-JPL-Caltech

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Check out our Github!

